Kubernetes and Istio

Comprehensive Technical Documentation

Generated: June 2026
Version 1.0

8 - Istio

Istio is an open-source, independent service mesh that provides the fundamental capabilities needed to run a distributed microservice architecture successfully. It acts as a dedicated infrastructure layer sitting on top of your applications, controlling how different parts of an application share data with one another.

> ❓ ***Why is it useful?***
> *The primary advantage of Istio is that it delivers **critical operational capabilities** without requiring any **changes to your application code**. Instead of developers manually programming libraries for routing, encryption, or logging into every single microservice, Istio handles these concerns transparently at the platform layer.*

Key capabilities:

- **Traffic Management**: Allows you to set up dynamic routing policies, load balancing, and traffic splitting (e.g., for canary deployments or A/B testing) based on various HTTP/TCP criteria.
- **Security**: Secures service-to-service communication by providing identity verification, strong access controls, authorization policies, and automated encryption.
- **Observability**: Generates detailed telemetry, distributed tracing, and logging for all traffic within the mesh, allowing you to visualize your architecture through tools like Prometheus and Grafana.

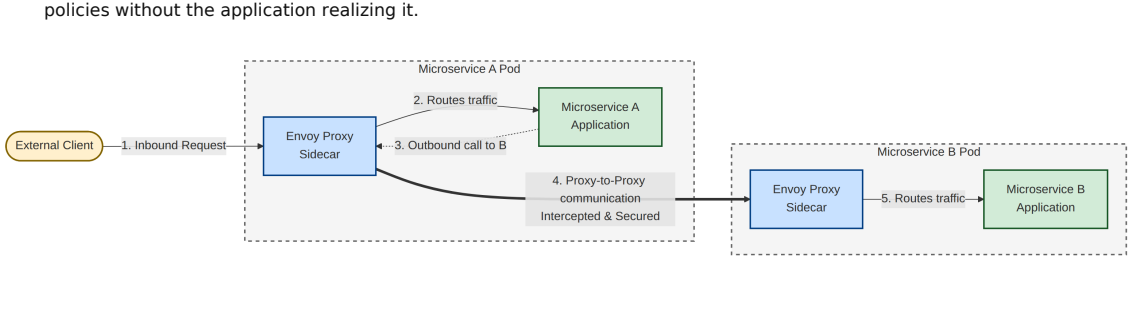
Istio architecture

Istio's architecture is split into two primary layers: the **Data Plane** and the **Control Plane**.

1. The Data Plane

The data plane consists of a network of intelligent proxies based on the open-source Envoy proxy.

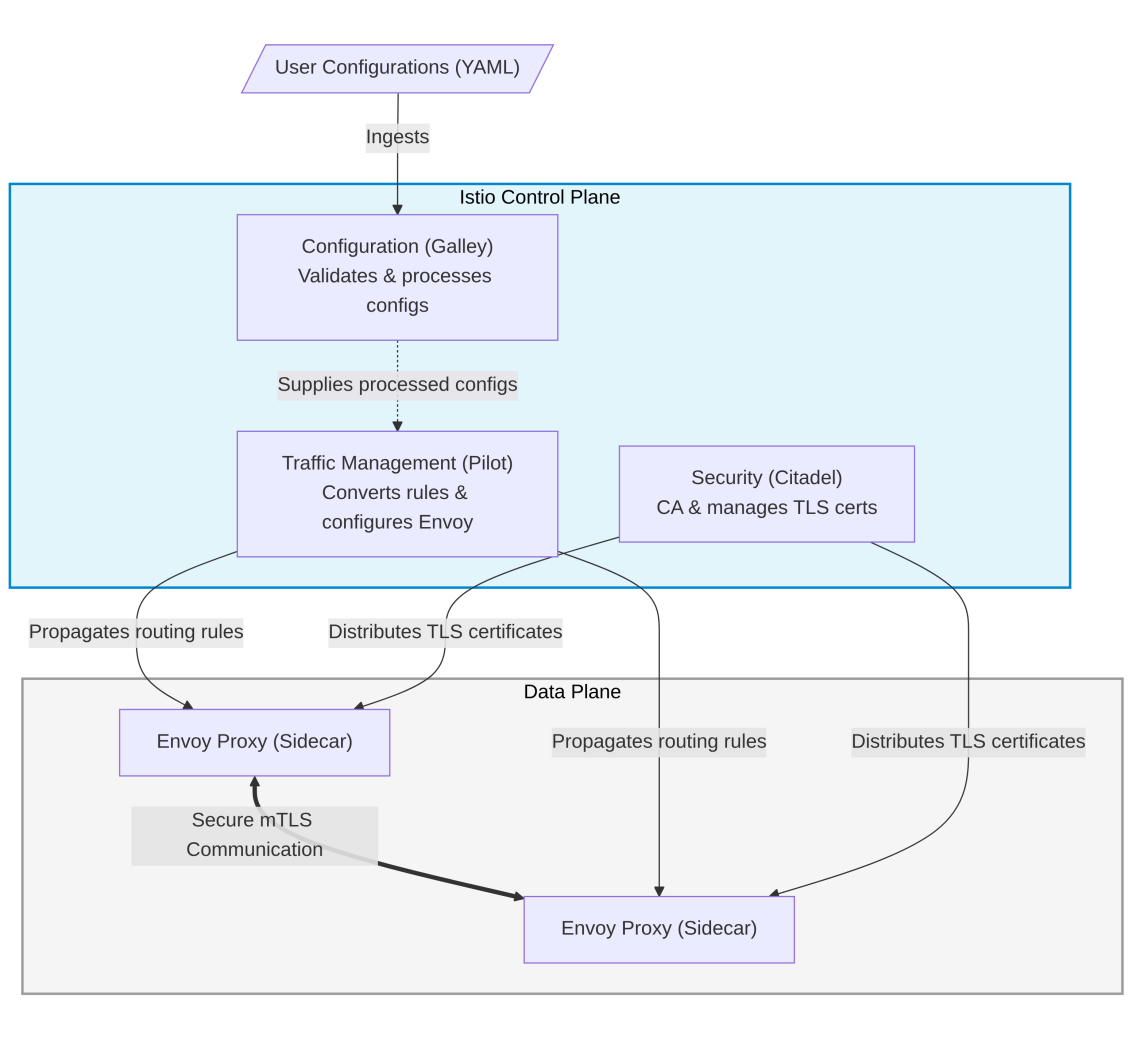
- Instead of exposing your microservice pods directly to the network, an Envoy proxy is deployed alongside your application inside the same pod as a sidecar container.
- The sidecar sits on the frontline: it intercepts all inbound and outbound network traffic for the microservice, executing policies without the application realizing it.



2. The Control Plane

The control plane manages and configures the proxies in the data plane to route traffic and enforce policies. It consists of several core components:

- **Traffic Management (formerly Pilot)**: Converts high-level routing rules into Envoy-specific configurations and propagates them to the sidecars.
- **Security (formerly Citadel)**: Acts as a Certificate Authority (CA), generating and managing TLS certificates to enable secure mutual TLS (mTLS) communication between microservices.
- **Configuration (formerly Galley)**: Validates, processes, and ingests user-provided configurations (like YAML files) into the Istio system.


User Configurations (YAML)
Ingests
Istio Control Plane
Configuration (Galley) Validates & processes configs
Supplies processed configs
Traffic Management (Pilot) Converts rules & configures Envoy
Security (Citadel) CA & manages TLS certs
Propagates routing rules
Distributes TLS certificates
Data Plane
Envoy Proxy (Sidecar)
Propagates routing rules
Distributes TLS certificates
Secure mTLS Communication
Envoy Proxy (Sidecar)

`istioctl` definition and setup

`istioctl` is the dedicated command-line configuration utility for Istio. It allows operators to debug the service mesh, verify installation status, and interact directly with the control plane.

Step-by-Step Installation:

- **Download the Release**: Run the official script to download the binary files and sample configurations (such as the bookinfo application) directly into your workspace:

```
curl -L https://istio.io/downloadIstio | sh -
```

- **Navigate to the Directory**: Move into the newly created folder (the version may vary based on the latest stable release downloaded):

```
cd istio-*
```

- **Add the Binary location to Your Path**: To make `istioctl` globally accessible in your terminal, move it to your local binary folder:

```
export PATH=$PWD/bin:$PATH
```

- **Verify the Host Installation**: Confirm the utility is active and ready:

```
istioctl --help
```

Installing Istio on k3d

Follow the guide at Istio k3d setup.

Prerequisite: Install Helm

To install `helm` follow the guide on `helm` website.

On Ubuntu:

```
HELM_BUILDKITE_APT_KEY_ID="DDF78C3E6EBB2D2CC223C95C62BA89D07698DBC6"

sudo apt-get install curl gpg apt-transport-https --yes

curl -fsSL https://packages.buildkite.com/helm-linux/helm-debian/gpgkey > "${TMPDIR:-/tmp}/helm.gpg"

# Ensure that the key ID matches to prevent a repository compromise from establishing an attacker controlled key
if [ "$(gpg --show-keys --with-colons "${TMPDIR:-/tmp}/helm.gpg" | awk -F: '$1 == "fpr" {print $10}' | head -n 1)" != "${HELM_BUILDKITE_APT_KEY_ID}" ]; then echo "ERROR: Unexpected Helm APT key ID: potential key compromise"; exit 1; fi

cat "${TMPDIR:-/tmp}/helm.gpg" | gpg --dearmor | sudo tee /usr/share/keyrings/helm.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/helm.gpg] https://packages.buildkite.com/helm-linux/helm-debian/any/ any main" | sudo tee /etc/apt/sources.list.d/helm-stable-debian.list

sudo apt-get update
sudo apt-get install helm
```

Istio installation

1. Create cluster with the following command

```
k3d cluster create --api-port 6550 --agents 2 --k3s-arg '--disable=traefik@server:*' istio-cluster
```

2. Install `istio` with `k3d`

```
helm install istio-base istio/base -n istio-system --set defaultRevision=default --wait --create-namespace
```

3. Install **Istio Discovery**

```
helm install istiod istio/istiod -n istio-system --wait
```

4. Verify installation of `istiod`

```
helm status istiod -n istio-system
```

5. Enable automatic sidecar injection

```
kubectl label namespace default istio-injection=enabled
```

6. Determine the Docker Network Subnet

```
docker network inspect k3d-istio-cluster | grep Subnet
```

7. Install `MetalLB` with Helm (see MetalLB website):

```
helm repo add metallb https://metallb.github.io/metallb
helm repo update
helm install metallb metallb/metallb -n metallb-system --wait --create-namespace
```

> ❓ ***Why installing MetalLB?***
> *"Vanilla" `k3d` does not emulate the scenario where Istio excels: **complex cloud environments**.*
> *It is necessary to install a `LoadBalancer` provider on the cluster for best cloud environment emulation.*

8. Create a file named `metallb-config.yaml`. **Make sure the addresses fall within the Docker subnet you found in Step 6**.

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: first-pool
  namespace: metallb-system
spec:
  addresses:
  - 172.20.0.200-172.20.0.250 # Update this range to match your k3d subnet!
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: homework
  namespace: metallb-system
```

9. Apply configuration

```
kubectl apply -f metallb-config.yaml
```

10. Install **Istio Ingress**

```
helm install istio-ingressgateway istio/gateway -n istio-ingress --wait --create-namespace
```

Run the Grafana dashboard on the cluster

Istio allows to easily instrument the cluster. You can install on the cluster Prometheus and Grafana with a default setup for monitoring cluster activities.

```
# 0. Move to Istio folder
cd istio-*

# 1. Install Prometheus
kubectl apply -f samples/addons/prometheus.yaml

# 2. Install Grafana
kubectl apply -f samples/addons/grafana.yaml
```

Now you can launch Grafana dashboard with

```
istioctl dashboard grafana
```

> ⚠️ ***Warning***
> *The `samples/addons/` YAMLs are strictly meant for **development, testing, and demonstration purposes**. They are not secured or optimized for production workloads.*
>
> *If you plan to run this cluster in production, the officially recommended path is to install Grafana and Prometheus using their own official Helm charts (like the kube-prometheus-stack), and then configure them to scrape your Istio endpoints.*